

Shaine Wei

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EDUCATION

Duke University, Pratt School of Engineering

Durham, NC

B.S.E. in Biomedical & Mechanical Engineering, Minor in Mathematics

May 2029 (Expected)

- BME Design Fellow | Dean's List | GPA: 3.87/4.00
- Relevant Coursework:** Quantitative Physiology & Biostatistical Applications, Molecular Biology, Organic Chemistry I, Linear Algebra, Advanced Multivariable Calculus, Ordinary and Partial Differential Equations, Data Structures & Algorithms, Engineering Innovation, Fundamentals of Electrical and Computer Engineering

EXPERIENCE

Undergraduate Research Assistant

Sept. 2026 – Present

EBC Laboratory, Duke University

Durham, NC

- Analyzing human biological samples for air pollution exposure biomarkers using mass spectrometry and chromatography
- Contributing to research linking environmental toxicant exposure to adverse human health outcomes

Co-Director & Tech Team Lead

Sept. 2025 – Present

HackDuke, Duke University

Durham, NC

- Lead HackDuke's executive team in organizing Duke University's annual hackathon, coordinating logistics for 200+ attendees
- Lead development of HackDuke's full-stack event platform using React, Python, and PostgreSQL, overseeing website reliability
- Mentor new developers through technical workshops, code reviews, and hands-on guidance in full-stack web development

Technical Communications Assistant

Sep. 2025 – Present

Mechanical Engineering & Materials Science, Duke University

Durham, NC

- Lead production of the bi-semester undergraduate newsletter in collaboration with faculty and research groups
- Translate technical research topics into accessible written content for a broad student audience
- Support departmental communications and event documentation

PROJECTS

RoboSub AUV – Claw Mechanism (Duke Robotics) | *SolidWorks, Mechanical Design*

Sept. 2026 – Present

- Designing a submersible claw end-effector for Duke Robotics' new AUV competing in the international RoboSub competition
- Engineering a gripping mechanism to autonomously grasp and manipulate objects during underwater mission tasks
- Collaborate within a mechanical team to take the claw from concept through CAD modeling, fabrication, and integration

Electric Vehicle Drivetrain Optimization (Duke Electric Vehicles) | *Mechanical Design, Drivetrain*

Sept. 2026 – Present

- Optimizing the drivetrain for Duke Electric Vehicles' new vehicle as a member of the mechanical subteam
- Analyzing brake disc and gearbox alignment to inform a redesigned, more tightly integrated drivetrain layout

Rube Goldberg Machine (EGR121L) | *Arduino, Sensors, Mechanical Design*

Apr. 2026

- Built a multi-stage electromechanical Rube Goldberg course on a 15-person team using Arduino-controlled sensors
- Led design and integration of an Arduino-controlled elevator that returned the ball to the start for a second run
- Coordinated timing across subteams to deliver reliable, cohesive performance of the full system

Sunflower Shelling Machine (EGR121L) | *SolidWorks, 3D Printing*

Feb. 2026

- Designed and fabricated a human-operated shelling machine achieving 2nd highest performance score in a 30+ student class
- Engineered a fully 3D-printed kernel/shell separation system with removable sifting tray and integrated collection chamber
- Modeled assemblies in SolidWorks and integrated a gear-driven dual-roller compression mechanism to minimize kernel damage
- Conducted quantitative testing of seed geometry and crushing force to inform channel spacing and roller cut dimensions

Yeast Genome Cloning Project (BIO201L) | *Molecular Biology*

Jan. – Apr. 2026

- Extracted genomic DNA from yeast cells and performed enzymatic digestion using RNase and restriction enzymes
- Constructed recombinant plasmids via ligation; applied antibiotic selection and screening to isolate successful clones
- Performed PCR and gel electrophoresis to amplify and analyze DNA fragments

TECHNICAL SKILLS

Programming: Python, Java, R, HTML/CSS, JavaScript, React, Git

CAD & Fabrication: SolidWorks, Fusion 360, Inventor, Onshape, 3D printing, laser cutting, woodworking

Lab Techniques: PCR, gel electrophoresis, immunohistochemistry, microscopy, mass spectrometry, chromatography

Data & Tools: R/Python data analysis, Excel, LaTeX, Arduino, microcontrollers

Languages: English (fluent), Mandarin Chinese (conversational)

ACTIVITIES & ORGANIZATIONS

American Society of Mechanical Engineers (ASME) | Biomedical Engineering Society (BMES) | DTech | Duke Electric Vehicles | Duke Robotics Club | HackDuke | Society of Women Engineers (SWE)